清华大学热能工程系 2005 级“燃烧学”期中考试试卷（A 卷）

Middle Term Examination of Combustion (A)

考试时间(Date)： 2008 年 4 月 25 日 班级(Class)：_____

学号(Student No.)：_____ 姓名(Name)：_____

Instructions(提示):

1 请将你的学生证放在桌子的右上角（Please put your Photo student ID on the desk）

2 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）

3 试卷共有11题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有5页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are 11 questions on this exam with 5 pages. You have up to 120 minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5 强烈推荐你先读一下所有的问题，并对回答问题的时间进行按排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

问题（problems）

1. 要组成 1 摩尔的 C_8H_{18}-空气混合物，当Φ=2/3 时，需要多少空气量？如果这一混合物进行燃烧，请给出燃烧后的产物的组成（分别按质量分数和摩尔分数来表示），假

设燃烧为完全燃烧。（How much air is needed for the preparation of a C_8H_{18}-air mixture with an equivalence ratio of Φ=2/3, assuming 1 mol of mixture. If this mixture burned, could you give the composition of the products in terms of mass fraction and mole fraction）.[10 分]

Solution：When Φ=2/3, the reaction can be expressed as：

$$C_8H_{18}+\frac{3}{2}(8+\frac{18}{4})(O_2+3.76N_2)\rightarrow 8CO_2+9H_2O+\frac{1}{2}(8+\frac{18}{4})O_2+\frac{3}{2}(8+\frac{18}{4})3.76N_2,$$

i.e. $C_8H_{18}+18.75(O_2+3.76N_2)\rightarrow 8CO_2+9H_2O+6.25O_2+70.5N_2,$

thus, for 1mol mixture，the air needed is：$\frac{18.75\times 4.76}{1+18.75\times 4.76}=0.9889$ mol

the composition of the products in terms of mole fraction is：

CO_2：$\frac{8}{8+9+6.25+70.5}=0.0853$

H_2O：$\frac{9}{8+9+6.25+70.5}=0.0960$

O_2：$\frac{6.25}{8+9+6.25+70.5}=0.0667$

N_2：$\frac{70.5}{8+9+6.25+70.5}=0.7520$，

And the composition of the products in terms of mass fraction is：

CO_2：$\frac{8\times 44}{8\times 44+9\times 18+6.25\times 32+70.5\times 28}=0.1309$

H_2O：$\frac{9\times 18}{8\times 44+9\times 18+6.25\times 32+70.5\times 28}=0.0603$

O_2：$\frac{6.25\times 32}{8\times 44+9\times 18+6.25\times 32+70.5\times 28}=0.0744$

N_2：$\frac{70.5\times 28}{8\times 44+9\times 18+6.25\times 32+70.5\times 28}=0.7344$

2. 假设在化学当量比条件下，计算 C_3H_8 和 O_2 燃烧的理论燃烧温度，产物有 50%的回流，并假设产物分解（H，O，OH,…）等可以忽略。即产物仅考虑为水和二氧化碳。（且燃料温度为 298K，氧气浓度为 398K，回流的烟气的温度为 448K，理想气体，压力为 1atm）Calculate the adiabatic flame temperature for the stoichiometric combustion of gaseous C_3H_8 with O_2 and with products recirculation(50%). The occurrence of dissociation products(H,O,OH,…)should be neglected; i.e.,water and carbon dioxide are the only products.(T_{Fuel}=298K, T_{Oxygen}=398K,$T_{recirculation}$=448K, ideal gas, p=1 atm).[12 分]

Solution：in stoichiometric condition, the reaction can be expressed as：

$C_3H_8+5O_2\rightarrow 3CO_2+4H_2O$

with 50% of products recirculation, the final reaction can be expressed as：

$C_3H_8+5O_2+3CO_2+4H_2O\rightarrow 6CO_2+8H_2O$，

According to the first law, when the pressure is constant, the enthalpy of the reactants and the products are the same.

T_{Fuel}=298K, T_{Oxygen}=398K，$T_{recirculation}$=448K，assume the adiabatic flame temperature is 3100K，the constant specific heats evaluated at 1700K，thus：

$$H_{react}=1\times(-103847)+5\times100\times30.21+3\times(-393546+\frac{41.276+44.569}{2}\times150)+4\times(-241845+\frac{34.437+35.337}{2}\times150)$$

$$H_{prod}=6\times[-393546+59.316\times(T-298)]+8\times[-241845+48.901\times(T-298)]$$

and： $H_{prod}=H_{react}$

we get：T=3108K，

It is consistent with our assumption, so the adiabatic flame temperature is 3108K.

3. 假设对化学反应 $H_2O+CO=CO_2+H_2$ ，其在 1100K 和 p=1 大气压下的平衡常数 K=1，试计算在 at p=8 atm，T=1100K 下的平衡常数 K。 Suppose for the (global) reaction $H_2O+CO=CO_2+H_2$ the equilibrium constant K=1 at 1100K and p=1 atm. Estimate K at p=8 atm , T=1100K?[6 分]

Solution: The equilibrium constant can be expressed as $K=\exp(-\Delta G_T^0/R_uT)$. For one reaction, ΔG_T^0 is a function which relates to temperature only. That is to say, it has nothing to do with the pressure. So, K is the same if the temperature is the same, i.e. K =1 at p=8 atm, T=1100K.

4. 下述反应中哪些是基元反应，哪些不是？哪些又是没有更多的信息是无法确定是否是基元反应？（Which of the following reactions are elementary ones, which not; for which reactions such a decision cannot be made without additional information:）[10 分]

1	**H_2+Br_2**	→	**2HBr**
2	**$2H_2+O_2$**	→	**$2H_2O$**

3	**$O+H_2$**	→	**$OH+H$**
4	**$2O+CH_3$**	→	**$CO+OH+2H$**
5	**$H+O_2$**	→	**$OH+O$**
6	**N_2+O_2**	→	**$2NO$**
7	**$H+H+M$**	→	**H_2+M**
8	**$NO+NO+O_2$**	→	**NO_2+NO_2**
9	**C_3H_8+M**	→	**$C_2H_5+CH_3+M$**

并对每个反应说明你的理由（Please give the arguments for your decision in each case!）

Solution：Elementary reaction is reaction that can occur in one collision. For these reactions, the elementary reactions are:

3，only one bond is broken, and one bond is formed.

5，only one bond is broken, and one bond is formed.

7，only one bond is formed, and M transfers the energy.

9，only one bond is broken, and M transfers the energy.

The reactions which are not elementary reactions are as follows：

1，two bonds are broken, two bonds are formed, and it is hard to satisfy the proper orientation between the colliding partners.

2，three bonds are broken, four bonds are formed, and it is hard to satisfy the proper orientation between the colliding partners.

4，three bonds are broken, two bonds are formed.

6，two bonds are broken, two bonds are formed, and it is hard to satisfy the proper orientation between the colliding partners.

Reaction which needs additional information is：8，for only one bond is broken, two bonds are formed, in high temperature and pressure this reaction may be a elementary reaction.

5. 在各种工业品的制造中都需要加入阻燃剂，一般采用卤素类元素作为阻燃剂，试从你学习的化学动力学知识分析其阻燃原理。（In most of the industrial products the fire

retardants should be added. Generally haloids are used as fire retardants. Could you give the explanation of the principle of the fire proof with the knowledge of chemical kinetics）[5 分]

Solution：Haloids can collide with the radicals and form a stable product. And radicals are very important in the combustion reactions. So, if the concentration of the radicals is reduced, the combustion process can be very slow. Therefore, haloids are used as fire retardants.

6. 一个详细的甲烷燃烧化学动力学要涉及到几百个，有人提出一个 14 步的化学反应作为其化学动力学简化机理 (A detail chemical mechanism of the CH_4 combustion includes several hundreds reactions. A reduced mechanism including only 14 steps is proposed to describe this mechanism)[15 分]：

Ⅰ $2O \Rightarrow O_2$

Ⅱ $O + H_2O \Rightarrow 2OH$

Ⅲ $O + CH_2O \Rightarrow H + OH + CO$

Ⅳ $O_2 + C_2H_2 \Rightarrow H_2 + 2CO$

Ⅴ $O_2 + CO \Rightarrow O + CO_2$

Ⅵ $O_2 + CH_2O \Rightarrow 2OH + CO$

Ⅶ $CH_2O \Rightarrow H_2 + CO$

Ⅷ $2H + CH_2O \Rightarrow OH + CH_3$

Ⅸ $C_2H_4 \Rightarrow H_2 + C_2H_2$

Ⅹ $OH + C_2H_2 \Rightarrow H + HCCOH$

Ⅺ $O + CO + CH_2O \Rightarrow O_2 + CH_2CO$

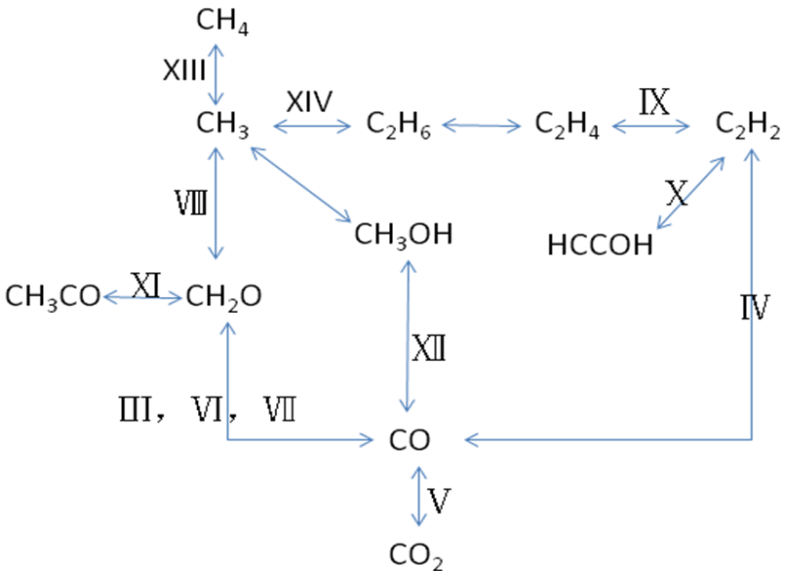
Ⅻ $O + H_2 + CO + H_2O \Rightarrow O_2 + CH_3OH$

ⅩⅢ $CH_3 + H_2O \Rightarrow OH + CH_4$

ⅩⅣ $2CH_3 \Rightarrow C_2H_6$

请你分析此简化反应可能的适用范围，并且分析一下这 14 步反应是如何获得的及每一步代表了甲烷氧化的何种关键步骤。（Please decide the possible range of this simplified mechanism. How do you think this mechanism come from? Please point out that which key step of every reaction of the mechanism represent in the oxidation of the methane.）

Solution: C_2H_6 and CH_3OH appear in the reactions. So, it is the reactions in the low temperature conditions (T<1500K). These reactions can be got according to the main reactions or composition in the low temperature conditions. The main process of the oxidation of methane is as the following figure：



Thus,

Reaction 1 and 2 provide radicals for the reactions.

Reaction 13 is the initial oxidation of CH_4.

Reaction 8 stands for the process that CH_3 oxidizes to the important medial product CH_2O.

Reaction 3，6 and 7 present different ways CH_2O oxidizes to CO.

Reaction 14 is the character of low temperature condition. Then C_2H_6 can oxidize to C_2H_2 through reaction 9, then CO through reaction 4.

Reaction 5 presents the formation of CO_2 by the oxidation of CO.

Reaction 11 presents a competitive reaction of the process of the formation of CO through CH_2O，Reaction 10 presents a competitive reaction of the process of the formation of CO through C_2H_2.

7. 对于动量传递方程 $\frac{\partial(r\rho v_x v_x)}{\partial x}+\frac{\partial(r\rho v_x v_r)}{\partial r}=\frac{\partial}{\partial r}\left(r\mu\frac{\partial v_x}{\partial r}\right)-r\frac{\partial P}{\partial x}+\rho g_x r$，其中左边两项都是对流项(亦特称惯性项)；右边第一项为扩散项(亦特称粘性项)，第三项为重力项。尝试：
(1)详细推导表征惯性项/粘性项、惯性项/重力项的无量纲数 [4 分];
(2)给出定义这两个无量纲数的具体名字(2 分)。

（As for the momentum conserved equation $\frac{\partial(r\rho v_x v_x)}{\partial x}+\frac{\partial(r\rho v_x v_r)}{\partial r}=\frac{\partial}{\partial r}\left(r\mu\frac{\partial v_x}{\partial r}\right)-r\frac{\partial P}{\partial x}+\rho g_x r$，the left hand of equation is convective term that is also called as inertia term, while the first and third terms of the right hand of equation are diffusive term (namely viscous term) and

gravity term, respectively. Please try:

(1) to deduce the dimensionless number of inertia/viscous as well as that of inertia/gravity;

(2) to give the name of these two dimensionless numbers.）

Solution：

(1) Firstly, the momentum conserved equation equals the following equation by considering the continuity equation,

$$\rho v_x r\frac{\partial v_x}{\partial x}+\rho v_r\frac{\partial (rv_x)}{\partial r}=\frac{\partial}{\partial r}\left(r\mu\frac{\partial v_x}{\partial r}\right)-r\frac{\partial P}{\partial x}+\rho g_x r$$

if we set length scale L, velocity scale V_0, then we have,

$$r^*=r/L;\ x^*=x/L;\ v_r^*=v_r/L;\ v_x^*=v_x/L;$$

Then we get

$$\frac{\rho V_0^2 L}{L}v_x^* r^*\frac{\partial v_x^*}{\partial x^*}+\frac{\rho V_0^2 L}{L}v_r^*\frac{\partial (r^* v_r^*)}{\partial r^*}=\frac{\mu L V_0}{L^2}\frac{\partial}{\partial r^*}\left(r^*\frac{\partial v_x^*}{\partial r^*}\right)-Lr^*\frac{\partial P}{\partial x}+\rho g_x L r^*$$

Thus, the dimensionless number of inertia/viscous term is deduced as,

$$\frac{\frac{\rho V_0^2 L}{L}}{\frac{L\mu V_0}{L^2}}=\frac{\rho V_0 L}{\mu}=\frac{V_0 L}{\nu}.$$

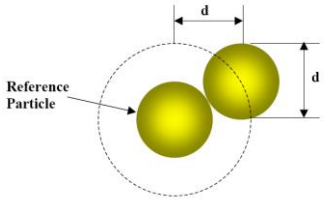
And the dimensionless number of inertia/gravity is deduced as,

$$\frac{\frac{\rho V_0^2 L}{L}}{\rho g_x L}=\frac{V_0^2}{g_x L}$$

(2) We set the dimensionless numbers of inertia/viscous term and inertia/gravity as Reynolds number and Froude number.

$$\mathrm{Re}=\frac{V_0 L}{\nu};\qquad Fr=\frac{V_0^2}{g_x L}$$

8. 如果 Z_{AB} 指的是单位时间、单位体积内 A 分子和 B 分子碰撞次数，则根据下式 $Z_{AB}=n_A n_B d_{AB}^2\pi(\frac{8k_B T}{\pi m_{AB}})^{1/2}$，试给出：(1)式子右边每一个符号代表的意义；(2)根据下图试说明这个式子成立的物理基础是什么？**[6 分]**（We define Z_{AB} as the collisions between molecule A and B per unit volume per unit time. According to the equation of $Z_{AB}=n_A n_B d_{AB}^2\pi(\frac{8k_B T}{\pi m_{AB}})^{1/2}$, please try: (1) to give out the detailed meaning of each symbol in the equation; (2) to explain the physical basis supporting this equation by considering the figure given below.）



Solution:

(1) $Z_{AB} = n_A n_B d_{AB}^2 \pi (\frac{8k_B T}{\pi m_{AB}})^{1/2}$ is valid by considering each symbol as

Z_{AB} – the collisions between A and B per unit time per unit volume, also collision frequency between A and B per unit volume;

n_A – the number of molecule A per unit volume

n_B – the number of molecule B per unit volume

d_{AB} – diameter the collision volume that is given as $d_{AB} = (d_A + d_B)/2$;

m_{AB} – the reduced mass where m_A and m_B are the masses of species A and B, which is defined as $m_{AB} = m_A m_B /(m_A + m_B)$;

k_B – Boltzmann constant

T – Absolute temperature

$(\frac{8k_B T}{\pi m_{AB}})^{1/2}$ – the mean molecular velocity (speed).

(2) If we focus on a single molecule of species A, only the molecule B that passes the spherical surface, πd_{AB}^2, within a short time Δt has the opportunity to collide with A. That is, only molecule B within a differential volume $\pi d_{AB}^2 \Delta l = \pi d_{AB}^2 (\bar{v}\Delta t)$ has the chance to collide with A. If the number density of molecule B is n_B, the we can get the collisions of all molecules B with one molecule A per unit time as $n_B \pi d_{AB}^2 \bar{v}$. Considering the number density of molecule A is n_A, then we finally obtain that the collisions between species A and B per unit time per unit volume is $n_A n_B \pi d_{AB}^2 \bar{v}$. That is, the equation of $Z_{AB} = n_A n_B d_{AB}^2 \pi (\frac{8k_B T}{\pi m_{AB}})^{1/2}$ is obtained.

9. 当空气掠过一个温度稍高的水面时，在水面上方和主气流之间会形成一个缓慢的、富含水蒸气的气体边界层，这个边界层可以认为是一种气体附着在液体表面的“滞止薄膜”；水蒸气蒸发速率可以采用 Stefan 流问题来求解。在一特定实验条件下，得到液面的蒸发速率是 6×10^{-4}kg/m^{2}s；液面处水蒸气的质量分数(Yi)为 0.0218，而在主流中水蒸气的质量分数是 0.0052；温度 298K；空气和水蒸气的摩尔质量分别是 28.85 和 18.02。 **[12 分]**

试求：

(1)如果质量扩散系数($D_{H2O\text{-}air}$,)为 2×10^{-5} m^2/s，试计算这个连接水面和气体主流的“滞

止薄膜”的厚度；

(2)如果质量扩散系数随着气体混合物中水蒸气组分而变化，即 $D_{H2O\text{-}air}=2\times10^{-5}\times(1+10Y_{H2O})$ m^2/s，试计算此时“滞止薄膜”的厚度。

（When air blows over a warm-water surface, a boundary layer of retarded, warmed and moistened air forms above the surface. This boundary layer can also be treated as “stagnation film” of gas adhering to the liquid surface; the rate of vaporization was calculated using the Stefan-flow equations. In a particular experimental situation, the rate of vaporization from the water surface is found to be $6\times10^{-4}kg/m^2s$; the mass fraction of water vapour at the water surface is 0.0218, and the mass fraction of water vapour in the bulk of the air stream is 0.0052. The temperature is 298K, the molar weights of air and water vapour are 28.85 and 18.02, respectively.

(1) If binary diffusivity, $D_{H2O\text{-}air}$, is 2×10^{-5} m^2/s, please calculate the thickness of the equivalent “stagnation film” of the air;

(2) If the binary diffusivity, $D_{H2O\text{-}air}$, is $2\times10^{-5}\times(1+10Y_{H2O})$ m^2/s, then calculate the thickness of “stagnation film” of the air. ）

Solution:

(1) Since the configuration give above represents the Stefan problem, we apply the equation as below,

$$\dot{m}''_{H2O}=\frac{\rho D_{H2O-air}}{L}\ln\left[\frac{1-Y_{H2O,\infty}}{1-Y_{H2O,i}}\right]$$

Since $Y_{H2O,\infty}=0.0052$ and $Y_{H2O,i}=0.0052$ are known, the mixture molecular weights at interface and in the bulk flow are then,

$$MW_{mix,i}=\frac{1}{Y_{H2O,i}/MW_{H2O}+(1-Y_{H2O,i})/MW_{air}}=\frac{1}{(0.0218/18.02+(1-0.0218)/28.85)}=28.477$$

$$MW_{mix,\infty}=\frac{1}{Y_{H2O,i}/MW_{H2O}+(1-Y_{H2O,i})/MW_{air}}=\frac{1}{(0.0052/18.02+(1-0.0052)/28.85)}=28.760$$

Then, $\overline{MW}_{mix}=(28.760+28.477)/2=28.618$. And then,

$$\bar{\rho}=\frac{101325}{(8315/28.62)298}=1.170\ kg/m^3$$

Then the thickness of the “stagnation film” can be calculated by,

$$L=\frac{\rho D_{H2O-air}}{\dot{m}''_{H2O}}\ln\left[\frac{1-Y_{H2O,\infty}}{1-Y_{H2O,i}}\right]=\frac{1.170\cdot2\times10^{-5}}{6\times10^{-4}}\ln\left[\frac{1-0.0052}{1-0.0218}\right]=6.56\times10^{-4}(m)$$

(2) If $D_{H2O-air}$ is not a constant, but can be described as $2\times10^{-5}\times(1+10Y_{H2O})$. Then,

$$\dot{m}''_{H2O}=Y_{H2O}\dot{m}''_{H2O}-\rho D_0(1+10Y_{H2O})\frac{dY_{H2O}}{dx}$$

$$\text{then}\quad \frac{\dot{m}''_{H2O}}{\rho D_0}dx=-\frac{(1+10Y_{H2O})}{1-Y_{H2O}}dY_{H2O}=10dY_{H2O}+11\frac{dY_{H2O}}{Y_{H2O}-1}$$

After integration, we get:

$$\frac{\dot{m}''_{H2O}}{\rho D_0} L = 10(Y_{H2O,\infty} - Y_{H2O,i}) + 11\ln\left[\frac{1-Y_{H2O,\infty}}{1-Y_{H2O,i}}\right]$$

Then, $L = 7.45\times10^{-4}(m)$.

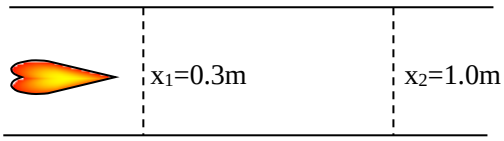
Note: some student also used the averaged diffusivity and can get the similar results. We still cut a little score since they don't verify the feasibility of this methodology. (注意：一些同学直接平均扩散系数也得到了很近似的结果，但是直接平均需要证明其方法和积分法的误差可行，为此酌情扣掉了 2-3 分)。

10. 假设天然气(CH_4)与空气混和物在柱塞流反应器中燃烧(稳态流动)，当量比是Φ=0.8，甲烷消耗量是 1.44kg/h(~0.0004kg/s)。在点 x_1=0.3m，温度为 1500K，随后温度线性下降，到点 x_2=1.0m 时，温度为 1150K, 且这两个点都处在火焰后的烟气区域内，反应器常压 1atm, 且反应器为直径 150mm 的圆管。**试求**：(1)点 x_1 处燃烧产物的平均密度； (2)从点 x_1=0.3m 到点 x_2=1.0m 之间距离内气相燃烧产物(烟气)停留时间。

[12 分] (提示) 应用公式 $dt_R/dx = 1/v_x$（Suppose that the natural gas (methane) burns with air at an air-fuel ratio (mass) of 0.8 in a plug-flow reactor (under a situation of steady-state and steady flow). The consumption of methane is 1.44kg/h(~0.001kg/s). There are two points in the postflame zone, x_1 (=0.3m) and x_2 (=1.0m), and the temperatures at x_1 and x_2 are 1500K and 1150K. The diameter of cylindrical reactor is 150mm. The reactor is at atmospheric ambience (1atm).

(1) Determine the averaged density of mixture at point x_1.

(2) Determine the residence time from point x_1 (=0.3m) to point x_2 (=1.0m).



Solution：

(1) Since the equivalence ratio is 0.8, *a*, representing the ratio of the number of O2 moles to the number of fuel moles, is calculated as,

$$a = \frac{x + y/4}{\phi} = \frac{2}{0.8} = 2.5$$

the lean or stoichiometric conditions becomes:

$$CH_4 + 2.5(O_2 + 3.76N_2) = CO_2 + 2H_2O + 0.5O_2 + (2.5\times3.76)N_2$$

Then, the mole fractions become:

$$\chi_{CO_2} = 0.078;\ \chi_{H2O} = 0.155;\ \chi_{O_2} = 0.039;\ \chi_{N_2} = 0.729$$

Then the mean mixture molecular weight and density are,

$$MW_{mix,i} = \sum \chi_i MW_i = 27.86$$

$$\bar{\rho}_{x=0.3m} = \bar{\rho}_0 = \frac{101325}{(8315/27.86)1500} = 0.226 \text{ kg/m}^3$$

(2) the mass air-fuel ratio, in general, is expressed as,

$$A/F = \frac{N_{air}}{N_{fuel}} \frac{MW_{air}}{MW_{CH4}} = 4.76a \frac{MW_{air}}{MW_{CH4}} = 4.76 \times 2.5 \times \frac{28.85}{16.04} = 21.404$$

Then the total mass flow rate in the tube is,

$$\dot{m} = (1 + A/F)\dot{m}_{CH4} = 0.0004 \cdot 22.404 = 0.00896 \quad (\text{kg/s})$$

Considering $\dot{m} = \rho v_x A$ and $\rho = \frac{PMW_{mix}}{R_u T}$,

Then,

$$\mathrm{dt_R}/\mathrm{dx} = 1/\mathrm{v_x} = \frac{\rho A}{\dot{m}} = \frac{PMW_{mix} A}{\dot{m} R_u T}$$

According to the given temperature profiles, we know,

$$T = 1500 + \frac{1150 - 1500}{1.0 - 0.3}(x - 0.3) = 1650 - 500x \quad (\text{K})$$

Thus,

$$\mathrm{dt_R}/\mathrm{dx} = \frac{PMW_{mix} A}{\dot{m} R_u} \frac{1}{1650 - 500x}$$

After the integration,

$$\mathrm{t_R} = \frac{PMW_{mix} A}{\dot{m} R_u} \frac{-1}{500} \ln(1650 - 500x)\Big|_{0.3}^{1.0}$$

$$= \frac{101325 \cdot 27.86}{0.00896 \cdot 8315} \frac{\ln(1500/1150)}{500} = 0.356 \text{ (s)}$$

(一些同学直接取平均温度可以得到结果 $\mathrm{t_R}$ = 0.360 (s) ,如果提供该简化方法的可行性证明，则给予满分，如果没有则根据计算结果酌情扣掉 2-4 分)。

11. 柱塞流反应器的能量守恒方程可写成 $\dot{m}'' \frac{d(h + v_x^2/2)}{dx} = -\frac{\dot{Q}'' P}{A}$ (其中 P 为反应器边界周长而 A 为反应器截面积)；而实际燃烧系统的能量守恒方程亦可写成 $\dot{m}'' \left(\frac{dh}{dx} + v_x \frac{dv_x}{dx} \right) = -\frac{d\dot{Q}_x''}{dx}$，请判断上述两个式子中 $\dot{Q}''$ 和 $\dot{Q}_x''$ 所代表的物理意义是否相同，并给出详细的解释。**[6 分]**（The energy conservative equation of plug-flow reactor is described as $\dot{m}'' \frac{d(h + v_x^2/2)}{dx} = -\frac{\dot{Q}'' P}{A}$ (where P and A represent local perimeter and cross-section area of the reactor); and the energy conservation of practical system is given as $\dot{m}'' \left(\frac{dh}{dx} + v_x \frac{dv_x}{dx} \right) = -\frac{d\dot{Q}_x''}{dx}$. Please judge whether the meanings of $\dot{Q}''$ and $\dot{Q}_x''$ in two equations are different or not! Please give the detailed explanation to your selection.）

Solution:

The meaning of the $\dot{Q}''$ and $\dot{Q}_x''$ in the two equations are greatly different. The $\dot{Q}''$ in the energy conserved equation of plug-flow reactor indicates the wall heat flux from the inner gases to the outside. Meanwhile, in simplified energy conservation equation, $\dot{Q}_x''$ represents the heat flux related to both conduction contribution of gas media and the species diffusion contribution. We still note two key points:

(1) The energy conservation in plug flow reactor doesn't consider the contribution of the diffusions, either by heat conduction or by species diffusion.

(2) the simplified energy conservation doesn't consider the wall heat flux, In practice, if we want to apply the energy conservation into the one-dimensional case, the heat exchange between the gas media and the outside through the wall should be considered.